

1 Detecting sparse biological invasions
2 through multi-scale remote sensing: a
3 drone–satellite framework for mapping
4 *Neltuma* in dryland ecosystems
5

6 **Abstract:**

7 Biological invasions are transforming dryland ecosystems worldwide, yet detecting sparse invasive
8 plants remains a fundamental Earth observation challenge because ecological targets often occur
9 below the grain of operational sensors. Here, we develop a hierarchical drone–satellite remote
10 sensing framework to detect and map invasive *Neltuma* (formerly *Prosopis*) across heterogeneous
11 dryland ecosystems, integrating UAV imagery, field observations and optical satellite data within a
12 linked multi-scale machine learning workflow.

13 Using drone surveys, we achieved classification accuracies of ~90% overall and ~87% for *Neltuma*
14 detection, while showing that fine-resolution observations alone are difficult to scale directly.
15 Integrating drone-derived products with satellite predictors improved landscape-scale classification
16 performance by 6.1% over satellite-only approaches (69.7% to 75.8%), while improving lower-bound
17 model performance even more strongly, raising minimum ensemble accuracy by 10 percentage points
18 (61% to 71%) and demonstrating greater robustness through cross-scale calibration. Detection
19 accuracy increased strongly with finer sensor grain, from ~50% using Sentinel-2 (10 m), to ~62% using
20 PlanetScope (3 m) and ~77% using WorldView-2 (1.6 m), identifying scale mismatch as a major
21 constraint on observing sparse ecological targets. Models evaluated using spatial block cross-
22 validation produced lower, but likely more realistic, accuracy estimates than random cross-validation,
23 highlighting the importance of accounting for spatial dependence when assessing model
24 performance.

25 Landscape-scale mapping showed *Neltuma* is already established across 16 km² (3.6%) of the study
26 area and is expanding across a further 356 km² (79.9%), highlighting substantial but uneven invasion
27 risk across rangelands important for people and wildlife. Beyond species mapping, our results show
28 how integrating fine-resolution ecological observations with satellite Earth observation via a
29 hierarchical drone–satellite remote sensing framework can help overcome scale mismatch, improving
30 early detection and management of sparse biological invasions. This multi-scale framework has wider

31 relevance for biodiversity surveillance, woody encroachment, degradation monitoring and dryland
32 early-warning systems.

33 **Keywords:**

34 Mesquite, Prosopis, Woody plant encroachment, Rangeland degradation, Ecological monitoring,
35 Environmental intelligence

1. Introduction

Rangeland degradation is intensifying across African drylands, threatening livelihoods, biodiversity and ecosystem function through interacting pressures from land-use change, invasive species and climate change. In dryland systems these pressures are often spatially heterogeneous and difficult to monitor, creating major challenges for environmental management and raising broader Earth observation questions about how to detect sparse ecological change across landscapes. In Botswana's rangelands, invasion by exotic plants, particularly the *Neltuma* genus (Figure 1), is a prominent example of this challenge (Dougill et al. 2016). Formerly named *Prosopis* (Hughes et al. 2022) and commonly known in Botswana as Mesquite or Sexanana, *Neltuma* species in southwest Botswana have formed a hybrid swarm comprising four pure lines (*N. chilensis*, *N. juliflora*, *N. velutina* and *N. glandulosa*) and at least nine hybrids and variants. In this study, *Neltuma* refers to all species, variants and hybrids (Muzila et al. 2010).

Neltuma invasion is a socially and ecologically consequential form of environmental change in African rangelands. As a globally significant invasive alien genus, *Neltuma* can alter ecosystem structure, compete strongly for water and transform savanna vegetation dynamics (Hussain et al. 2021). In South Africa, native woody species density, richness and diversity decline with increasing *Neltuma* encroachment (Shackleton et al. 2015), while canopy dieback and mortality have been recorded in keystone Kalahari savanna trees (Schachtschneider and February 2013). In southwest Botswana, local communities associate *Neltuma* spread with falling water tables and report broader impacts on health and livelihoods (Mosweu et al. 2013). Yet these invasions are complex, with reported benefits including dune stabilisation, windbreaks, livestock fodder and fuelwood, complicating management responses (Mosweu et al. 2013).

Effective management of *Neltuma* depends on spatial information that has largely been lacking. Botswana's strategy of control by utilisation recognises both the ecological threat and socio-economic complexity of invasion, and the *Integrated Mesquite (Prosopis Species) Management Strategy for*

61 *Botswana* (2022) calls for context-specific responses in affected communal areas. However, planning,
62 prioritisation and monitoring require reliable spatial information on invasion extent, density and
63 progression at both local and landscape scales. Such information has been difficult to obtain using
64 accessible mapping techniques.

65 Detecting sparse invasive plants across heterogeneous drylands remains a persistent scale-mismatch
66 problem in Earth observation. Measuring *Neltuma* distribution in southwest Botswana is particularly
67 challenging because the invasion often occurs at low fractional cover, spans diverse growth forms and
68 phenological stages, and is embedded within heterogeneous vegetation mosaics. Effective monitoring
69 therefore requires methods capable of detecting individual plants during early incursion while also
70 capturing low-density populations across broader landscapes (Müllerová et al. 2023). Satellite remote
71 sensing has proven effective for mapping some invasive plants (Robinson et al. 2016; Rajah et al.
72 2019; Rebelo et al. 2021a), and *Neltuma* has been mapped using WorldView-2, Sentinel-2 and
73 Landsat imagery (Adam, Mureriwa, and Newete 2017; Berg and Catharina 2013; Eckert et al. 2025;
74 Page et al. 2022), but satellite approaches remain constrained by spatial and spectral resolution when
75 ecological targets are sparse or patchy (McMillan et al. 2023). This can limit sensitivity precisely where
76 early detection is most important.

77 Integrating drone and satellite observations offers new opportunities to bridge scale gaps in
78 ecological monitoring. Drone-based remote sensing can capture ecological structure at resolutions
79 inaccessible to satellites and provides a valuable intermediary between field observations and
80 landscape-scale Earth observation (Alvarez-Taboada, Paredes, and Julián-Pelaz 2017; Baena et al.
81 2017; Kattenborn et al. 2019; Cunliffe, Brazier, and Anderson 2016). Combined with satellite imagery,
82 drone data create opportunities for multi-scale analytical frameworks that improve interpretation of
83 coarser imagery, strengthen ecological inference and support more operational environmental
84 monitoring. Such approaches can also help guide local control and eradication efforts, inform policy
85 and management planning, and enable monitoring of follow-up interventions through repeatable,

86 scalable observations. They may be particularly valuable for sparse invasions, where sensor grain
87 strongly influences detectability and where management often depends on linking fine-scale
88 ecological information with landscape-scale decision-making.

89 This exploratory study assessed whether *Neltuma* can be mapped in ways useful to ecologists, land
90 users and managers. We combined ground observations with remote sensing from drone and satellite
91 sensors to address four research questions: (1) Can drone-derived data reliably detect *Neltuma* at
92 local scales in the Kalahari, and which combinations of data and machine learning algorithms
93 maximise accuracy? (2) Can drone-derived data improve the performance of satellite-based
94 classifications of *Neltuma*? (3) What are the limitations of optical satellite data for detecting *Neltuma*
95 in the Kalahari? (4) What insights does remote sensing provide into the invasion dynamics of *Neltuma*
96 in the region?

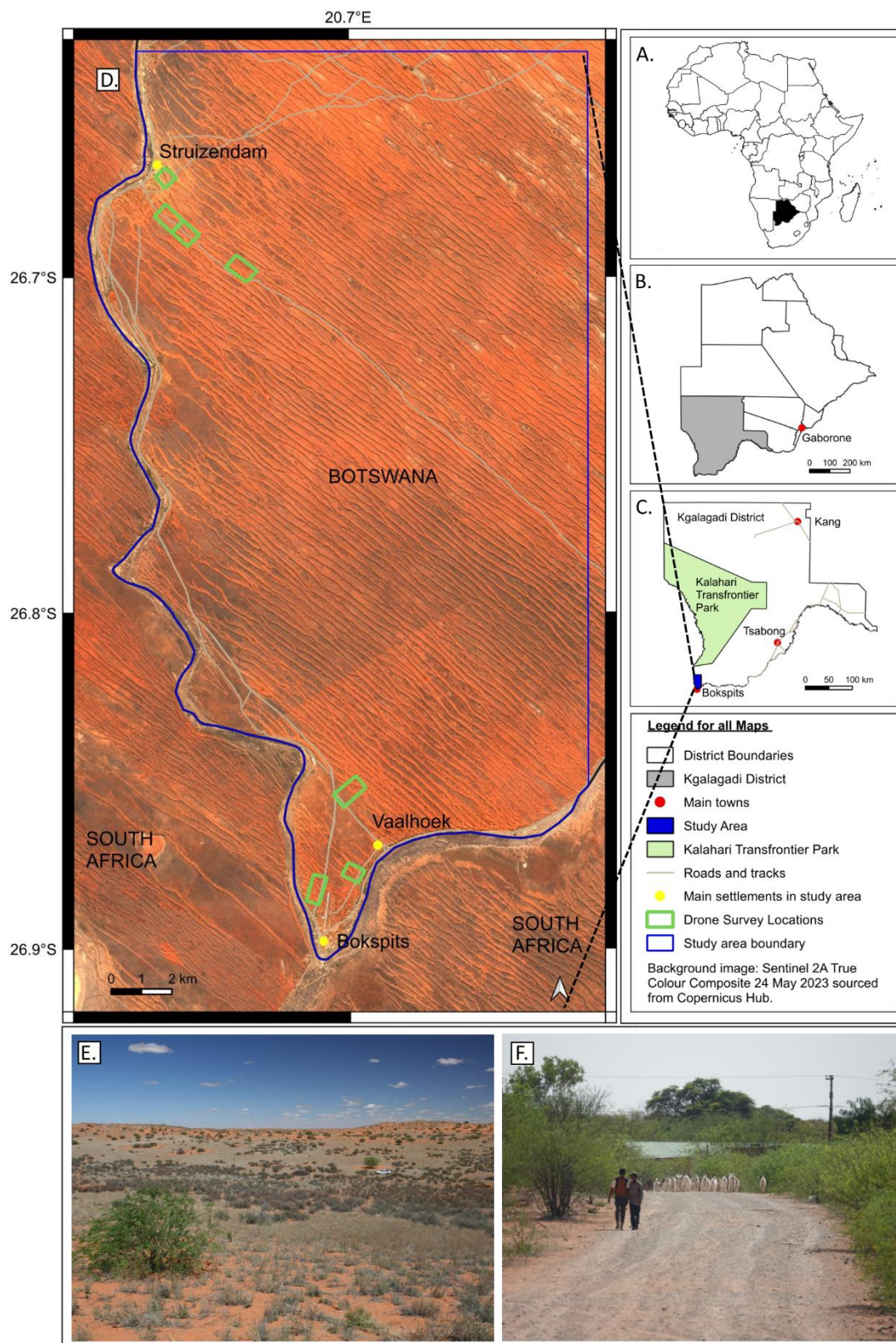
2. Methodology

2.1 Study system and research design

We used a hierarchical multi-scale Earth observation design to investigate sparse Neltuma invasion, integrating field observations with drone and optical satellite imagery. The study was structured to evaluate plant-scale detection, assess how fine-resolution observations could improve broader-scale satellite classification, quantify the effects of sensor grain on invasion detection, and characterise spatial patterns of invasion across the study area. Fine-resolution drone observations provided detailed ecological information for calibrating and evaluating satellite-based classifications, while satellite imagery enabled assessment of how detection performance changed as spatial resolution decreased. By linking these observation scales, the design allowed us to examine both the value of fine-resolution data for improving landscape-scale inference and the limitations introduced when sparse invasion patterns are observed at coarser spatial grains.

The study was conducted across approximately 445 km² of Kgalagadi rangelands in southwest Botswana surrounding the villages of Bokspits, Struizendam and Vaalhoek, bounded to the south by the Molopo and to the west by the Nossob ephemeral river valleys, which also mark the Botswana–South Africa border (Figure 1). The region comprises a semi-arid savanna system of deep sandy soils, variable grazing pressure and heterogeneous vegetation structured by NW–SE dune systems, dune valleys and dry river corridors. Vegetation is dominated by Kalahari Duneveld shrub-grassland, with dune crests supporting *Stipagrostis amabilis* and scattered trees such as *Vachellia haematoxylon* and *Boscia albitrunca*, while dune valleys contain grassland and low-density bush savanna with species including *Schmidtia kalahariensis*, *Stipagrostis uniplumis*, *Eragrostis lehmanniana* and *Aristida meridionalis*, interspersed with denser shrub patches of *Rhigozum trichotomum* and *Senegalia mellifera*. Dry river valleys support more open tree savanna including *V. erioloba* (Van Rooyen and Van Rooyen 2019). Communal pastoralism is the dominant land use, with continuous grazing by livestock shaping vegetation structure and disturbance regimes (Kgosikoma and Batisani 2014). Within this

122 dynamic dryland mosaic, *Neltuma*, introduced during sand dune stabilisation near Bokspits in 1980
123 (van der Poel and Timberlake 1980), has spread from initial planting sites into settlements and
124 surrounding savanna (Mosweu et al. 2013), occurring across a continuum from isolated individuals
125 and low-density populations to dense established stands, making it a suitable system for evaluating
126 scale-dependent detection.



127

128 Figure 1. Multi-scale remote sensing of sparse *Neltuma* invasion is situated within a heterogeneous

129 Kalahari dryland landscape. (A–C) Location of the study area within Africa, Botswana and the

Kgalagadi District. (D) The 445 km² study area surrounding Bokspits, Struizendam and Vaalhoek, showing Kalahari dune systems, settlements, drone survey sites and study area boundaries over Sentinel-2 imagery. (E) Representative Kalahari Duneveld vegetation illustrating the heterogeneous shrub–grassland mosaic in which sparse *Neltuma* occurs. (F) Dense *Neltuma* invasion near settlements and transport corridors.

2.2 Ground observations

Ground observations provided reference data for model training, independent validation and ecological interpretation across the multi-scale framework. Field surveys were designed to capture variation in *Neltuma* occurrence, density, growth form and associated land-cover classes across the invasion gradient, including isolated individuals, low-density populations and dense established stands, with emphasis on sparse occurrences that present the greatest detection challenge.

Field observations were collected between 1 September and 2 October 2022 (Table S2), during the late dry season when predominantly evergreen *Neltuma* contrasted strongly with largely quiescent or marcescent vegetation, reducing phenological mismatch with remotely sensed data. Surveys spanned all seven drone survey areas and recorded representative observations for the principal land-cover and vegetation classes used in classification (Tables S3 and S4).

Training and validation points were digitised in the field using QField v2.6.2 on GNSS-enabled tablets displaying orthomosaics generated from drone surveys processed on site in Pix4DMapper v4.8.0 (Pix4Dmapper 2022). Sampling points were selected so an 80 cm radius contained only a single land-cover class, with a minimum of 20 observations per class in each drone survey area. Spectral values associated with each field point were extracted using the R package *exactextractr* (Baston 2023), sampling pixels within 30 cm of point centres to calculate weighted averages of reflectance values; each field point represented approximately 80–105 image pixels depending on terrain and flight height.

Independent plant-scale validation surveys were conducted in two 100 × 200 m plots within drone survey areas, where the central location and maximum height of every *Neltuma* individual (n = 184) were recorded alongside representative observations of other vegetation classes and bare ground. These data were used to assess plant-level detection performance across growth forms and phenological states and to complement pixel-based accuracy assessment. Additional field observations were used to interpret ecological associations relevant to invasion dynamics, including frequent co-occurrence of *Neltuma* beneath native woody canopies. Together, these observations provided the ecological reference data underpinning classification development, validation and interpretation across scales.

2.3 Drone observations and image processing

Drone observations provided the fine-resolution ecological observation layer linking field measurements to landscape-scale satellite analysis. UAV surveys were designed to resolve individual plants at spatial grains beyond the capability of satellite sensors while generating ecological products for cross-scale calibration.

Drone surveys were conducted across seven sites spanning gradients in *Neltuma* density, geomorphology and vegetation structure near Bokspits and Struizendam (Figure 1D; Table S2), with site selection balancing ecological representativeness, community preferences and Botswana Civil Aviation Authority regulations. Surveys were undertaken between 1 September and 2 October 2022 during the late dry season to maximise spectral contrast between predominantly evergreen *Neltuma* and co-occurring species such as *R. trichotomum* and *S. mellifera* in largely quiescent or marcescent states. Flights were conducted under low-wind, clear-sky conditions to minimise artefacts in photogrammetric reconstruction, given evidence that canopy heights derived from drone photogrammetry can be sensitive to wind speed (Slade et al. 2025).

Imagery was acquired using a DJI Phantom 4 Multispectral (P4MS) with a six-camera array (Table S1), paired with a DJI D-RTK2 High Precision GNSS Mobile Station for positional control. Flight plans were

implemented in Ground Station Pro using nadir acquisition at 90–100 m above ground level, 80% forward and side overlap, and a 2.1 s image interval, yielding 5.8–6.7 cm ground sampling distance. Radiometric calibration followed empirical line methods using a 50% Zenith Lite Diffuse Reflectance Target, with calibration images collected pre- and post-flight and processed following established multispectral workflows (Assmann et al. 2018; Smith and Milton 1999). Multispectral imagery was processed in Pix4DMapper v4.8.0 to generate calibrated orthomosaics for each spectral band. RGB imagery was processed using structure-from-motion and multi-view stereopsis to generate dense 3D point clouds, from which Digital Surface Models and Digital Terrain Models were used to derive Canopy Height Models (CHM), later resampled to harmonise with multispectral data for classification analyses.

Drone-derived predictor sets included calibrated spectral bands, vegetation indices and structural variables, including CHM-derived metrics, evaluated in combination with alternative machine-learning algorithms to identify configurations maximising classification performance (Table S3). Drone imagery was used both to assess local-scale detectability of sparse invasions and to generate drone-derived products for cross-scale calibration in satellite classifications. Independent plant-level validation was used to assess detection performance before integrating these products into landscape-scale analyses.

2.4 Satellite observations and scale-mismatch analysis

Satellite observations were used to evaluate detection of sparse *Neltuma* invasions across contrasting sensor grains using WorldView-2 (1.6 m), PlanetScope (3 m) and Sentinel-2 (10 m), representing fine-, intermediate- and moderate-resolution optical Earth observation systems. Imagery was selected to align as closely as possible with field and drone surveys while minimising phenological mismatch (Table S5), providing a structured basis for assessing how sensor grain influences detectability.

Pre-processing included sensor-specific radiometric and atmospheric correction, after which predictor layers were assembled from spectral bands and vegetation indices (NDVI, SAVI, MSAVI2 and MTVI2; Table S6), selected to capture complementary variation in vegetation greenness and canopy signal under sparse vegetation and exposed-soil conditions (Rajah et al. 2019; Rebelo et al. 2021b; Francis et al. 2024; Slade et al. 2023). To support cross-scale calibration, drone-derived ecological products were co-registered with each satellite dataset and incorporated alongside satellite-only predictors in comparative classifications.

Because sparse training targets make classifications sensitive to positional error, satellite imagery was co-registered to drone data using maximum mutual information matching implemented through the `coregisterImages` function in *RStoolbox* (v0.3.0) (Leutner et al. 2022), correcting residual offsets among sensors (Rufin et al. 2021).

To test whether drone-derived classifications could inform satellite mapping, fractional vegetation cover was extracted from drone classifications to grids matching each satellite pixel footprint using *exactextractr* (Baston 2023). Because heterogeneous vegetation mosaics rarely produced uniform coarse pixels, we identified for each sensor grain the minimum cover threshold yielding ≥ 100 pixels per class (Table S7), then randomly resampled the master dataset to equalise class representation and reduce class imbalance.

Scale mismatch was assessed by comparing classification performance among sensors and between models with and without drone-derived calibration products, testing how detectability changed with increasing pixel size and whether fine-resolution ecological information improved sensitivity to low-cover invasions. These comparisons formed the basis for evaluating sensor-grain constraints and the contribution of multi-scale integration to landscape-scale invasion mapping.

2.5 Hierarchical multi-scale classification framework

We implemented a hierarchical multi-scale classification framework to integrate field, drone and satellite observations for detecting and mapping *Neltuma* across nested spatial scales (Figure 2). The framework comprised two linked machine-learning pipelines: a drone workflow generating fine-resolution classifications and drone-derived ecological products, and a satellite workflow evaluating sensor performance and the contribution of cross-scale calibration.

Drone classifications combined calibrated multispectral bands, vegetation indices and canopy height model predictors with field-derived training and validation data. To identify the best-performing classifier, we compared default RF, tuned RF, XGBoost (XGB), support vector machines (SVM) and ensemble (EN) models across four drone predictor stacks: (1) 5-band multispectral (Drone), (2) Drone + canopy height model (Drone+CHM), (3) Drone + CHM + NDVI, and (4) Drone + CHM + full vegetation index suite (Drone+CHM+All VI). Supervised classification was implemented in R using the *mlr3* framework (Lang et al. 2019), benchmarking models using 200-iteration spatial block cross-validation (SPCV) with Bayesian optimisation. We implemented stacked generalisation, in which cross-validated out-of-fold predictions from multiple base learners were used to train a meta-learner that generated the final classification, improving predictive robustness.

An analogous workflow was applied to WorldView-2, PlanetScope and Sentinel-2 using both satellite-only predictors and predictor sets augmented with drone-derived calibration products. Field observations were aligned to each sensor, with training points buffered by 0.8 m to generate labels for WorldView-2 classification. The same SPCV and optimisation framework was applied across sensors to identify optimal learner–predictor combinations.

Model performance was assessed from SPCV benchmarking using confusion-matrix-derived overall and class-level accuracies, with particular emphasis on sensitivity to *Neltuma* detection. This framework propagated observations across scales, evaluated sensor-grain effects and tested whether centimetre-scale ecological information strengthened landscape-scale inference.

2.6 Spatial validation and uncertainty assessment

Classification robustness and uncertainty were assessed using repeated accuracy metrics, block-based spatial cross-validation and independent plant-level validation. Across 200 model iterations, we quantified overall and class-level accuracies, with particular emphasis on sensitivity to sparse and low-cover *Neltuma* detections, using repeated spatially explicit evaluation to support robust uncertainty estimation (Meyer and Pebesma 2022; Milà et al. 2022). Independent validation using mapped *Neltuma* individuals from field plots was used to assess plant-level detection performance and omission and commission errors beyond pixel-based metrics.

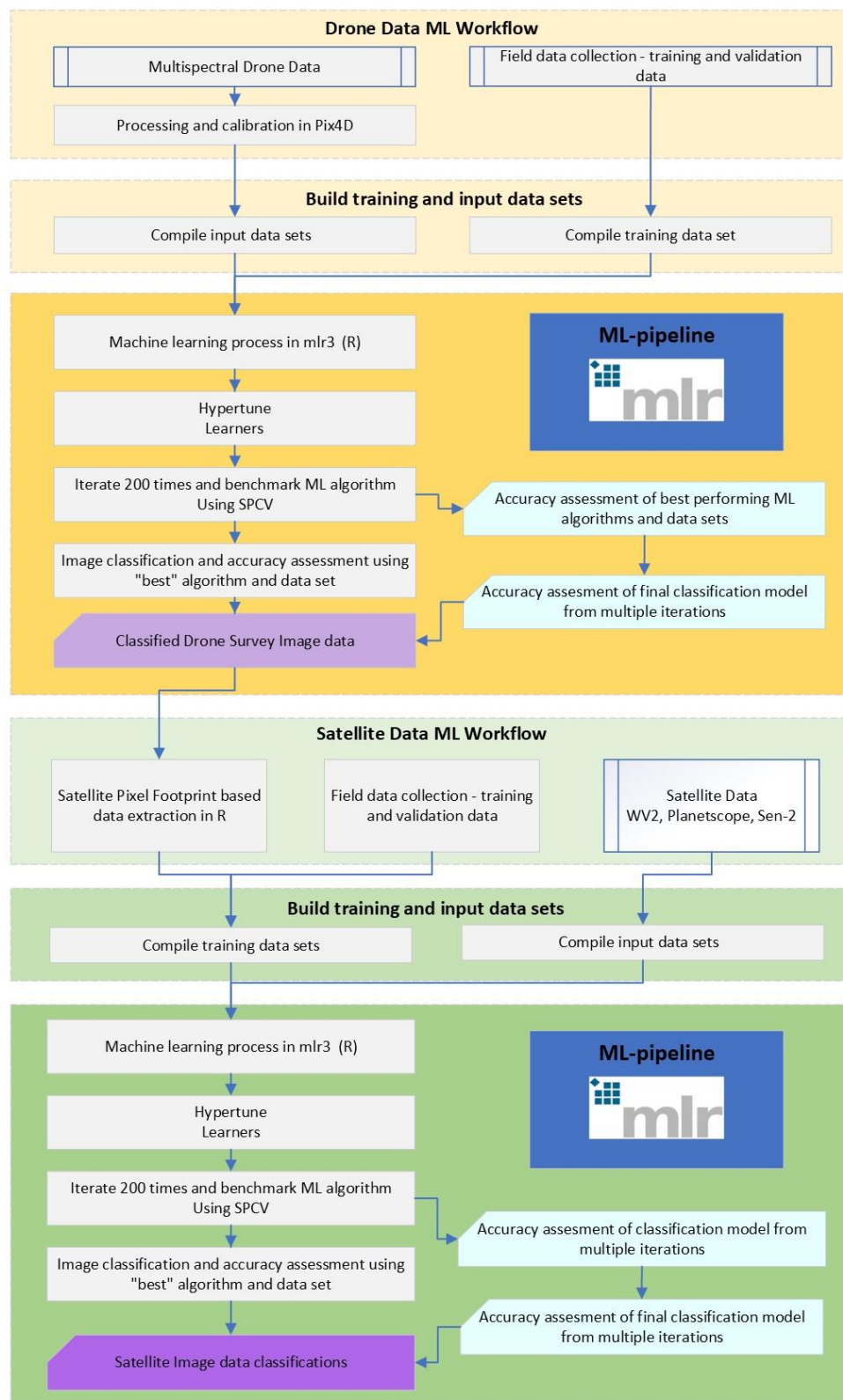
To evaluate predictive realism under landscape heterogeneity, we compared conventional and spatially explicit validation estimates, assessing how spatial structure influenced model performance and uncertainty. For satellite classifications, uncertainty associated with sensor grain and cross-scale calibration was assessed by comparing results among WorldView-2, PlanetScope and Sentinel-2, and between models with and without drone-derived predictors. Together, these analyses provided a robust basis for interpreting mapped invasion extent and invasion phase.

To support interpretation of species co-occurrence patterns, we assessed potential *Neltuma* interactions with *V. erioloba* by denoising final drone classifications using the sieve filter in *terra* (v1.7-37 Hijmans 2023), converting classified patches >1 m² to polygons, and analysing nearest-neighbour distances among woody species polygons using *sf* (Pebesma 2018; Pebesma and Bivand 2023). We visually verified results for *V. erioloba* against photointerpretation from the RGB orthomosaics.

2.7 Mapping invasion extent and invasion phase

We used the best-performing satellite classifications to generate landscape-scale maps of *Neltuma* invasion extent and invasion phase across the 445 km² study area. We quantified invasion extent from final classified outputs and classified *Neltuma* prevalence into invasion phases based on cover thresholds to aid land manager interpretation (Table S8; after Müllerová et al. 2023), distinguishing areas of advanced invasion from broader zones of active expansion and treating invasion as a dynamic landscape process rather than presence–absence alone.

275 We used the mapped invasion patterns to explore spatial variation in *Neltuma* occurrence and density
276 in relation to settlement proximity, geomorphic setting and vegetation structure. These exploratory
277 observations informed interpretation of invasion dynamics and potential management relevance
278 across the study system.



279

280 Figure 2. Hierarchical machine-learning workflows integrate drone and satellite observations for
 281 multi-scale classification of sparse *Neltuma* invasion. Parallel workflows compile predictor and training
 282 data, optimise and benchmark algorithms in mlr3 using spatial cross-validation across 200 iterations,

283 and generate final classifications with repeated accuracy assessment. Drone-derived products inform
284 satellite models through cross-scale calibration.

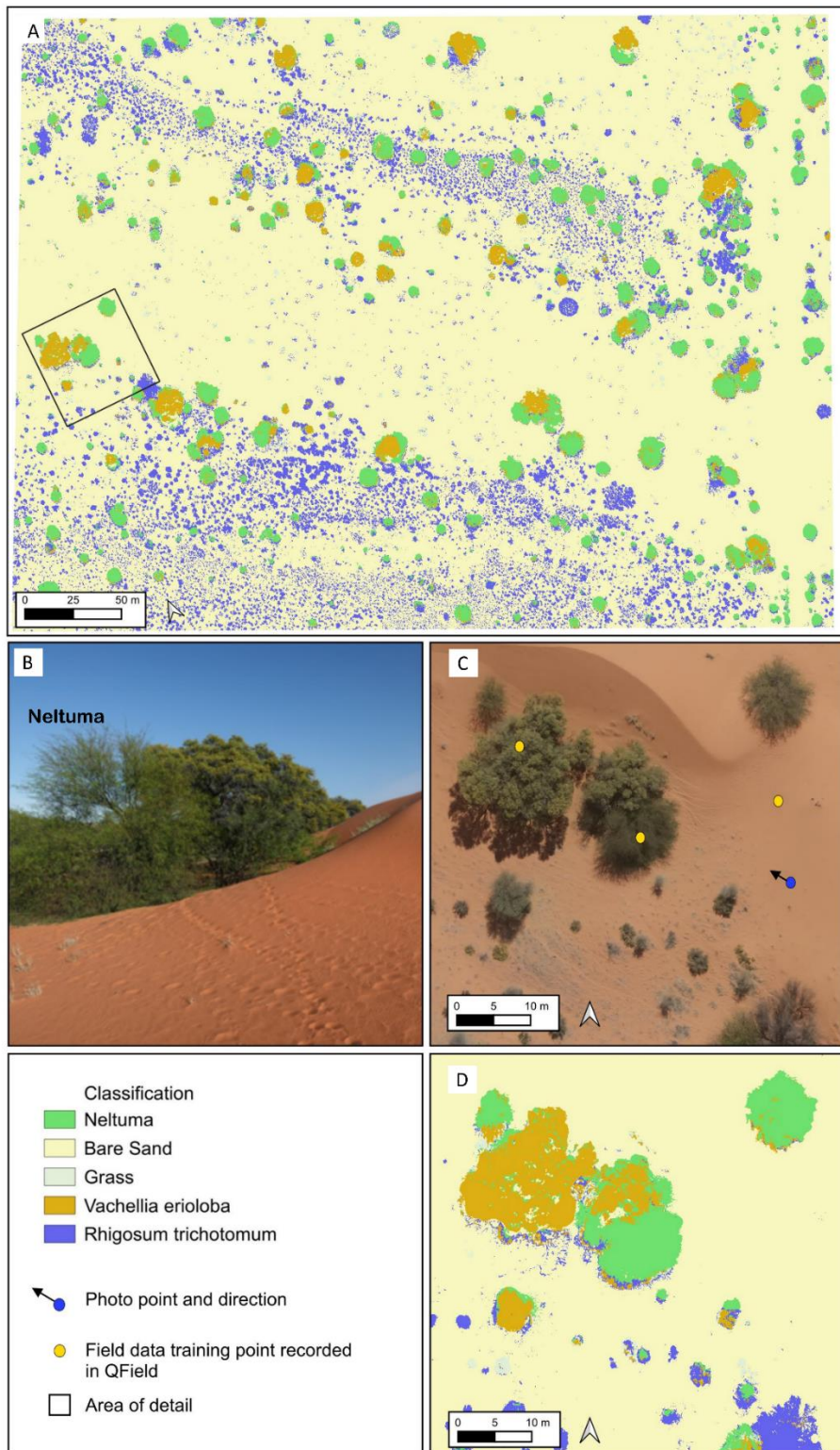
3. Results

3.1 Drone data enable accurate detection of sparse *Neltuma* at local scales in the Kalahari

Classifications across the seven drone survey areas achieved high performance, with mean overall accuracies of 85–90% against field observations. Final classification maps for all survey areas are shown in Figures S1–S7, with an example from Bokspits_1 in Figure 3. At centimetre grain, classifications resolved sparse *Neltuma* and associated vegetation effectively, although some misclassification occurred at the scale of individual branches, particularly along canopy margins and in shadowed areas where reflectance differed from the canopy as a whole.

Benchmarking across survey areas and predictor stacks showed consistently strong performance, with SVM and ensemble models achieving the highest mean overall accuracies of 91.3% and 90.3%, respectively (Figure 4A). The multispectral + CHM + NDVI predictor stack yielded the highest overall accuracy (Figure 4B), whereas *Neltuma*-specific accuracy was highest using multispectral data combined with CHM (87.2%; Figure 4C). Principal component analysis and band-level boxplots further indicated strong spectral separability between *Neltuma* and other vegetation classes (Figures S8–S9).

Most *Neltuma* plants were correctly classified at pixel level, although shaded canopy edges were occasionally confused with *R. trichotomum* and *V. erioloba* (Figure 3D). Independent plant-scale validation showed 84.7% detection accuracy (Table S9), with reliable detection across a range of plant heights and growth stages. Phenological variation, from leaf-off to flowering states, was the principal source of error. Misclassification occurred in 8.7% of cases, typically involving shaded or senescent plants, while 3.8% of *Neltuma* individuals were misclassified as *V. erioloba*, all in flower. Beyond classification performance, *Neltuma* occurred beneath or within 1 m of the canopy of 63% of *V. erioloba* trees (n = 192), indicating frequent co-occurrence consistent with native tree canopies acting as loci of invasion, with implications for interpreting invasion dynamics and targeting management.



308

309 Figure 3. Drone multispectral classification resolves sparse *Neltuma* and species associations at plant
 310 scales in heterogeneous Kalahari vegetation. (A) Example SVM classification from the Bokspits_1
 311 survey area showing fine-scale discrimination of *Neltuma*, *V. erioloba*, *R. trichotomum*, grass, and bare
 312 sand. (B) Field photograph illustrating *Neltuma* occurring beneath a *V. erioloba* canopy on a dune
 313 margin. (C) True-colour orthomosaic detail with field training points. (D) Corresponding classified

close-up showing species-level discrimination, including minor edge misclassification at canopy margins.

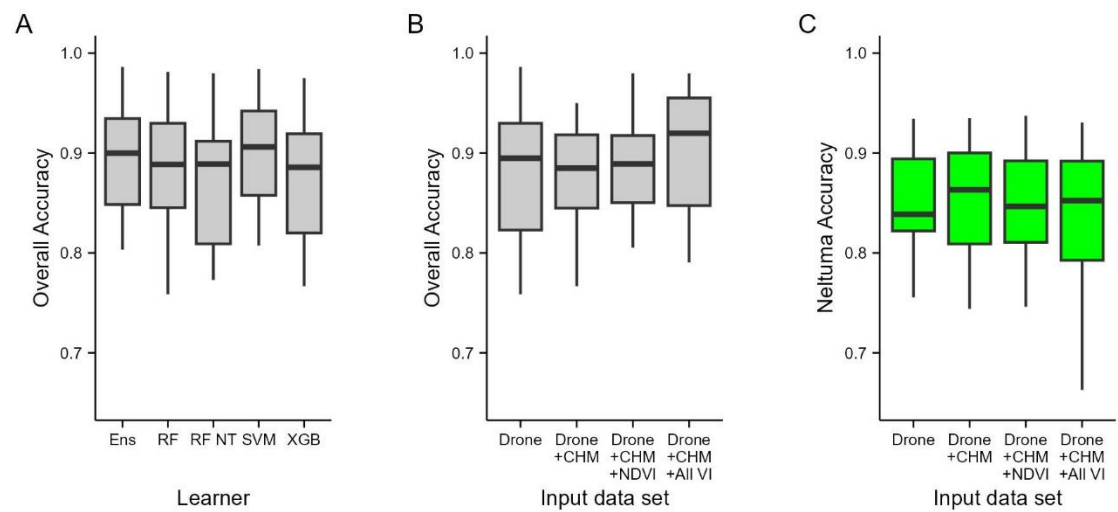


Figure 4. Drone multispectral classification achieved ~90% overall accuracy for vegetation mapping and reliable detection of sparse *Neltuma*. (A) Classification performance across five machine-learning algorithms over 200 spatial cross-validation iterations, with SVM achieving the highest mean overall accuracy. (B) Overall accuracy was highest using the multispectral, canopy height model (CHM) and NDVI (Drone = 5-band multispectral, ALL VI = combined vegetation indices) predictor stack. (C) *Neltuma*-specific accuracy was highest using multispectral data combined with canopy height information.

3.2 Cross-scale calibration with drone data improves satellite-derived classifications of *Neltuma*

Cross-scale calibration addressed a major constraint in satellite detection of *Neltuma*: its low fractional cover within coarse pixels. Across the seven drone survey areas, only 67 Sentinel-2 pixels (0.003%) contained > 75% *Neltuma* cover, showing that at 10 m grain *Neltuma* rarely dominates pixel reflectance (Figure 5). As pixel size increased, high-cover *Neltuma* pixels became progressively rarer, forcing classifications to rely largely on mixed-cover training data and illustrating a fundamental scale mismatch in detecting sparse invasions. Fine-resolution drone observations helped overcome this constraint by providing ecologically representative calibration data for satellite models.

Applied to WorldView-2 imagery, the *mlr3* workflow produced robust landscape-scale classifications, with mean overall accuracy of 75.8% (Figure 6). SVM and ensemble models again performed best, achieving mean accuracies of 75.9% and 75.1%, respectively. Incorporating drone-derived calibration

data improved predictive accuracy by 6.1%, from approximately 69.7% to 75.8%, while reducing variability and raising minimum ensemble accuracy by 10 percentage points, from approximately 61% to 71% (Figure 6). These results show that cross-scale calibration substantially strengthened satellite detection performance.

Independent comparison with the drone-derived classification revealed discrepancies that were not fully reflected by the satellite classification's internal validation metrics. The satellite classification overpredicted *Neltuma* extent by 24.8% relative to drone classifications (Table S10), while 34.6% of *Neltuma* plants identified in drone data were liable to be classified as *V. erioloba* or other woody vegetation in satellite imagery. These comparisons indicate greater classification error than suggested by internal training and validation statistics alone, while reinforcing both the value of drone-informed calibration and the importance of spatially explicit validation.

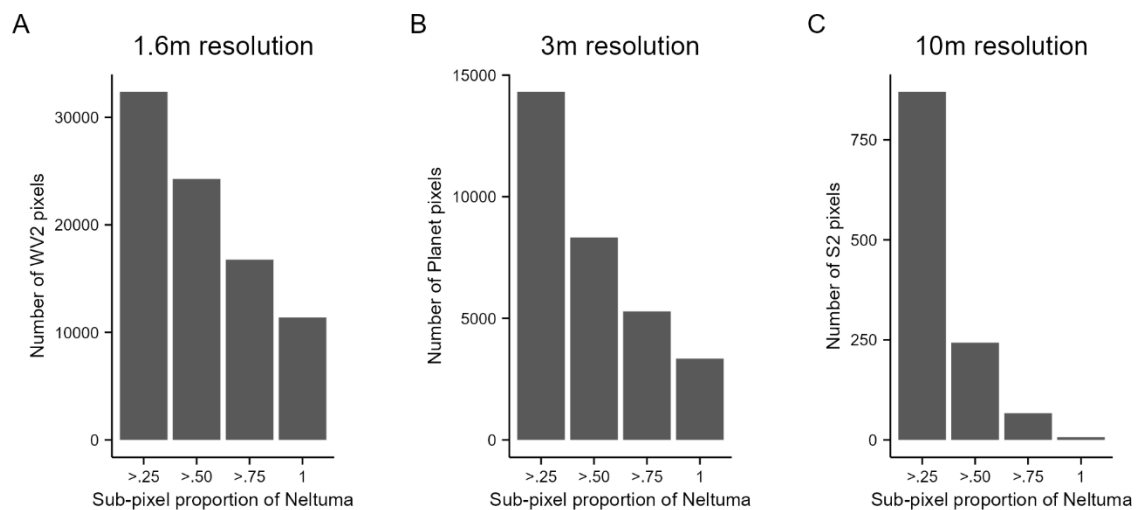
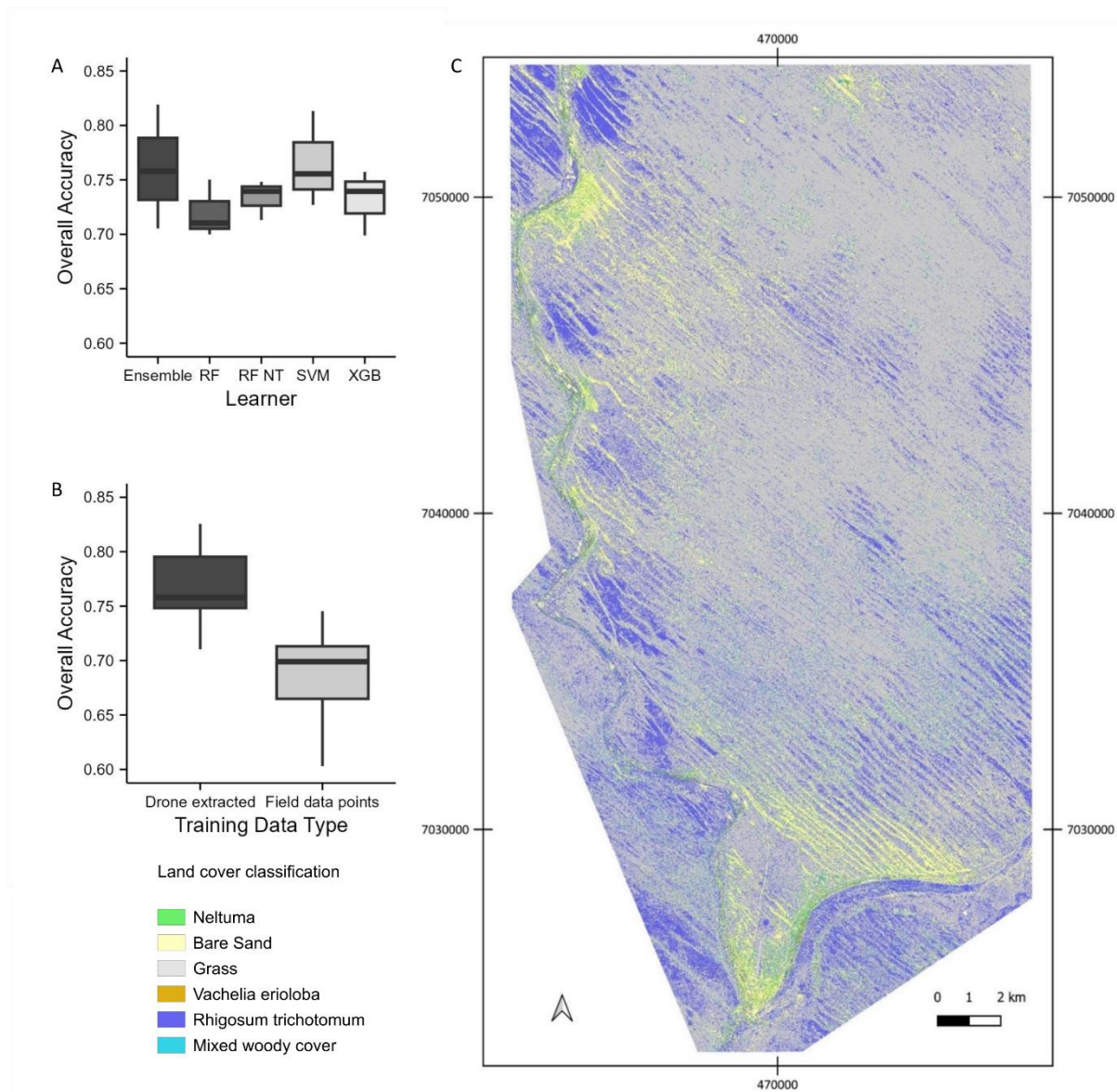


Figure 5. Coarser sensor grain sharply reduces the availability of high-*Neltuma*-cover pixels for training and detection. (A) Distribution of sub-pixel *Neltuma* cover within 1.6 m WorldView-2 pixels. (B) Corresponding distributions for 3 m PlanetScope pixels. (C) Corresponding distributions for 10 m Sentinel-2 pixels. Increasing pixel size reduces the frequency of high-cover pixels and illustrates the scale mismatch constraining detection of sparse invasions.

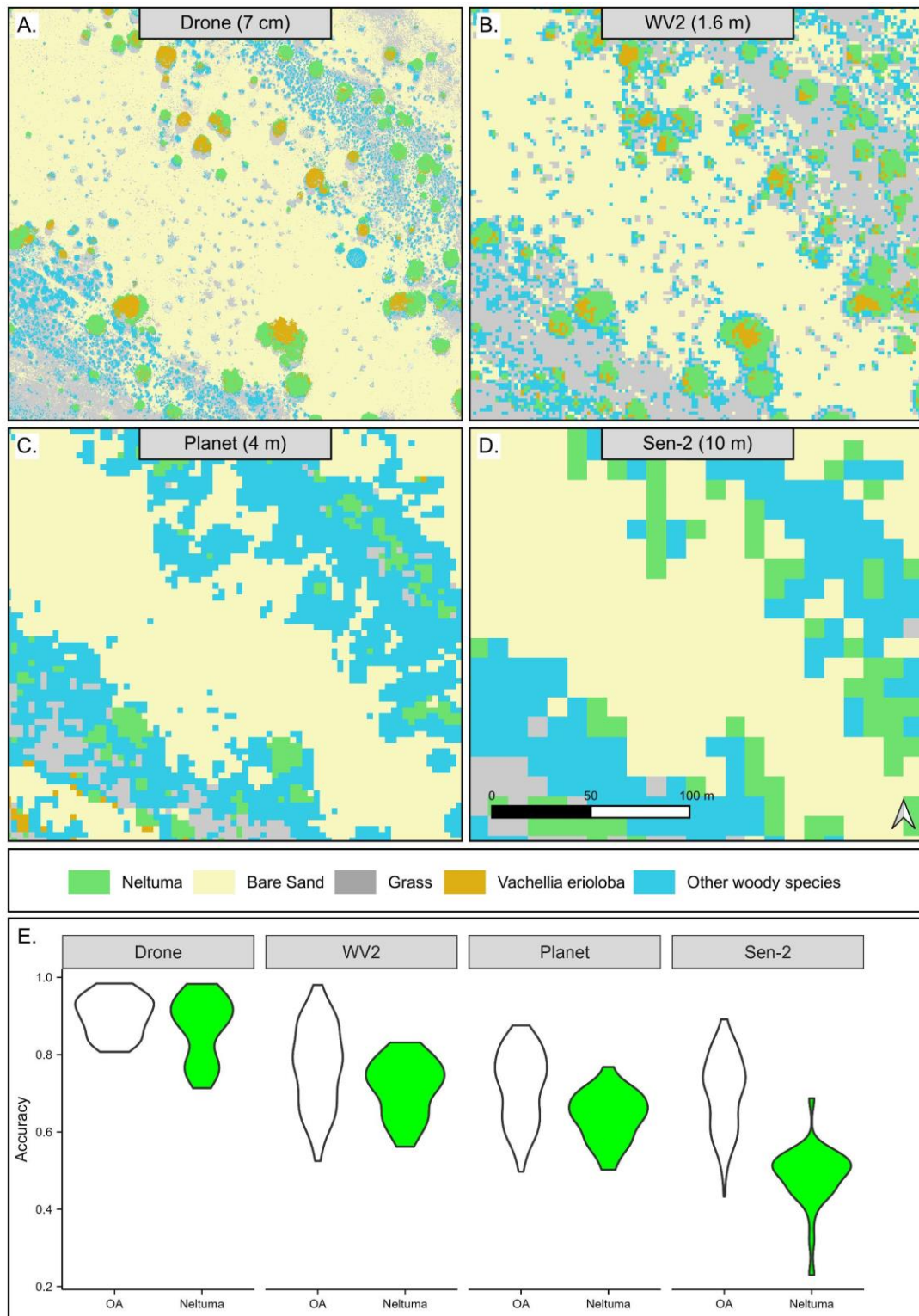


352

353 Figure 6. Cross-scale calibration with drone data increased satellite classification accuracy and
 354 improved landscape-scale mapping of sparse *Neltuma*. (A) Classification performance across five
 355 machine-learning algorithms over 200 spatial cross-validation iterations, with ensemble models
 356 achieving the highest overall accuracy. (B) Incorporating drone-derived calibration data increased
 357 WorldView-2 classification accuracy by 6.1% relative to field observations alone. (C) Final classification
 358 of WorldView-2 imagery showing spatially explicit *Neltuma* patterns across the 445 km² study area.

3.3 Detection of sparse *Neltuma* in the Kalahari is constrained by satellite image grain

Increasing sensor grain reduced *Neltuma* detectability, increased confusion with *V. erioloba* and other woody vegetation, and caused progressive loss of spatial detail in mapped invasion patterns (Figure 7). Although mean overall classification accuracy declined only slightly from PlanetScope to Sentinel-2 imagery, from 71.1% to 70.3%, *Neltuma*-specific accuracy declined much more sharply with increasing pixel size, falling to just 50.4% in the Sentinel-2 classification (Figure 7E). This divergence shows that conventional overall accuracy can mask substantial losses in sensitivity to sparse ecological targets.



366

367 Figure 7. Detectability of sparse *Neltuma* declines sharply as sensor grain becomes coarser.
 368 Classification outputs for the same area using (A) drone multispectral imagery (7 cm), (B) WorldView-
 369 2 (1.6 m), (C) PlanetScope (4 m), and (D) Sentinel-2 (10 m), illustrating progressive loss of spatial detail
 370 and increasing confusion among vegetation classes with coarser resolution. (E) Overall and *Neltuma*-
 371 specific accuracies from the best-performing algorithm, showing a disproportionately strong decline
 372 in *Neltuma* detection with increasing pixel size.

3.4 *Neltuma* invasion is widespread and expanding across the Kalahari study landscape

Neltuma invasion was widespread across the study area, with much of the landscape falling within initial incursion or expansion phases (Figures 6 and 8; Table 1). To support management interpretation, we simplified the multi-species WorldView-2 classification (Figure 6) into maps of *Neltuma* prevalence (Figure 8A) and invasion phase (Figure 8B; Table 1), highlighting broad low-density distributions alongside concentrated zones of advanced invasion. Although *Neltuma* occurred widely at relatively low cover, both invasion density and maturity increased in and around settlements and along roads, declining with distance from these features (Figures S10 and S11). This spatial distribution is consistent with field observations implicating livestock as a key vector of *Neltuma* spread.

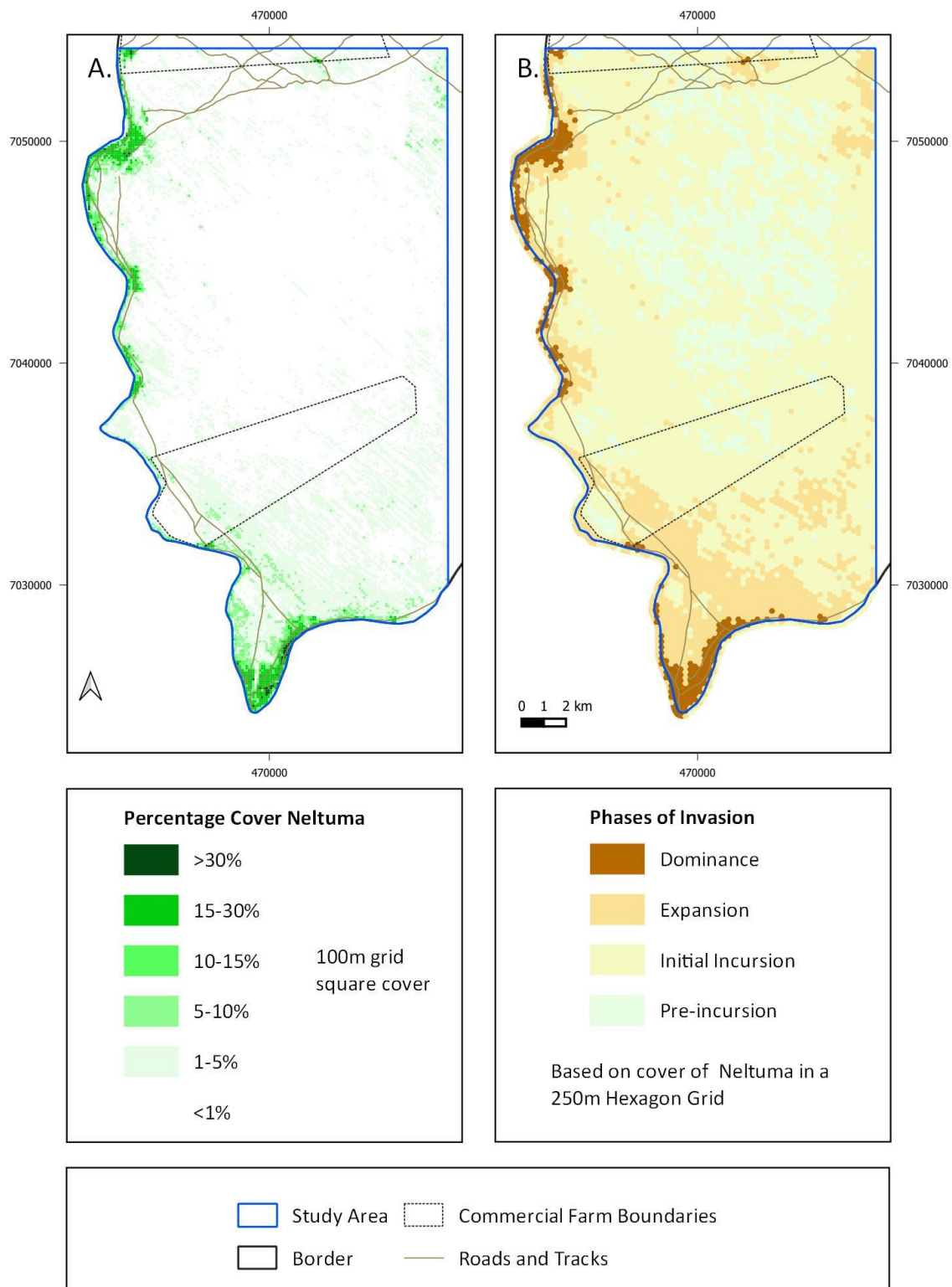


Figure 8. Spatial patterns of *Neltuma* prevalence reveal widespread invasion and extensive areas vulnerable to further spread. (A) Percentage cover of *Neltuma* mapped in 100 m grid cells from WorldView-2 classification. (B) Invasion phases derived from percentage cover in 250 m grid cells distinguish dominance from expansion and early incursion zones, supporting management prioritisation.

389

390 Table 1. Most of the study landscape is already affected by incipient or expanding *Neltuma* invasion.

Phase of Invasion	Percentage of Botswana study area (%)	Km ²
Dominance	3.6	16.01
Expansion	17.4	78.36
Initial Incursion	62.5	281.34
Pre-incursion	16.5	74.27

391

392 **4. Discussion**

393 Exotic invasive plants such as *Neltuma* threaten African rangelands by degrading ecosystem services
394 for people and wildlife, yet detecting sparse invasions remains challenging because invasion patterns
395 often occur below the grain of operational Earth observation systems. This study shows that sparse
396 invasion detection is fundamentally a scale-mismatch problem, but that integrating field observations,
397 drone imagery and satellite data within a hierarchical multi-scale framework can improve detection,
398 quantify sensor-grain constraints and support management-relevant mapping of invasion dynamics.
399 In the Kalahari, this approach revealed both the value of fine-resolution ecological information for
400 strengthening landscape-scale inference and the limitations of coarser satellite systems for detecting
401 low-cover invasions at expanding invasion fronts.

402 **4.1 Multispectral drone observations provide accurate plant-scale detection of sparse *Neltuma***

403 Multispectral drone observations provided accurate plant-scale detection of sparse *Neltuma*,
404 demonstrating the capacity of fine-resolution Earth observation to resolve invasion signals that are
405 often difficult to detect with coarser sensors (Slade et al. 2023). High classification performance
406 across diverse growth forms indicates that combining spectral and structural information can

discriminate *Neltuma* effectively within heterogeneous dryland vegetation, although detection was weaker for leaf-off individuals, highlighting phenology as a persistent source of uncertainty. Improved performance when vegetation indices were included reflects the value of phenologically sensitive predictors in dryland systems, while canopy height information reduced confusion with *V. erioloba* and other woody species by capturing structural differences not resolved by spectral information alone. These results were obtained using a pixel-based classification framework and therefore include some misclassification associated with canopy margins, mixed pixels and shadowed areas that may be reduced by object-based approaches incorporating spatial context.

Although all algorithms performed well, the marginally stronger performance of SVM, together with gains from ensemble modelling, suggests that predictor composition and robust model evaluation were more important than classifier choice alone. The block-based spatial cross-validation (SPCV) framework was particularly important because spatially structured observations can lead to overly optimistic estimates of predictive performance when conventional random train–test splits are used (Roberts et al. 2017). By repeatedly evaluating models across spatially distinct training and test partitions, our approach provides a more realistic assessment of how well classifications may generalise beyond the locations used for model development (Milà et al. 2022). Considering variation across 200 spatial cross-validation iterations, rather than mean accuracy alone, provided a more informative assessment of model stability across spatial contexts.

These findings demonstrate that drone observations as a powerful ecological observation layer for detecting sparse invasions and characterising invasion structure at local scales. While object-based approaches may further improve crown delineation and reduce local pixel-level errors, the pixel-based framework adopted here was sufficient to achieve accurate plant-scale detection and to support the subsequent cross-scale analyses. However, their principal value in this study lies not as a stand-alone monitoring solution, but as the fine-resolution foundation of a hierarchical multi-scale framework. Given the spatial extent of invasion in landscapes such as the 105,200 km² Kgalagadi

District, drone surveys alone do not provide operationally scalable monitoring, but they can supply the ecological detail and calibration needed to strengthen broader-scale satellite mapping.

4.2 Cross-scale calibration with drone data improves satellite classification of *Neltuma*

Cross-scale calibration with drone data improved satellite classification of *Neltuma*, demonstrating how fine-resolution ecological observations can strengthen inference at broader spatial scales. Incorporating drone-derived training information increased both classification accuracy and robustness in WorldView-2 analyses, suggesting that the improvement was not simply a consequence of increasing the number of training observations, but also reflected better representation of the spectral and structural variability associated with invasion targets and surrounding vegetation. This is particularly important for sparse invasions, where limited field samples may underrepresent the heterogeneous and mixed-cover conditions typical of real landscapes (Slade et al. 2023). These findings contribute to a small but growing body of work integrating observations across spatial scales, with geolocated fine-scale data from UAVs used to improve classification and prediction in agricultural applications (Maimaitijiang et al. 2020; Ogungbuyi et al. 2024), strengthen aboveground biomass estimation (Cheng et al. 2024; Kolluru et al. 2026), and support invasive plant mapping (Holden, Rebelo, and New 2021; Page et al. 2022). However, comparable hierarchical calibration frameworks remain rare in sparse invasive plant detection (Singh, Surasinghe, and Frazier 2024).

The benefits of cross-scale calibration should, however, be interpreted alongside uncertainty in transferring information between observation scales. The WorldView-2 classification achieved a mean overall accuracy of 75.8% following drone-derived calibration, but comparison with independent drone classifications revealed discrepancies not captured by internal validation. The satellite classification overpredicted *Neltuma* extent by 24.8%, while 34.6% of *Neltuma* plants identified in drone data were liable to be classified as *V. erioloba* or other woody vegetation. These differences demonstrate that improved cross-validated performance does not eliminate the challenges of mapping sparse targets at coarser sensor grains, and highlight the value of independent fine-

resolution observations for revealing ecologically meaningful errors not captured by conventional accuracy metrics. Future multi-scale invasion mapping should therefore move beyond assigning each coarse pixel to a single discrete class towards models that predict proportional cover of invasive species (Shiferaw et al. 2019; Frazier and Wang 2011). Such approaches could better capture uncertainty and spatial heterogeneity across observation scales, particularly where mixed pixels become increasingly prevalent and confidence in species presence varies continuously. Conformal prediction approaches offer one potential framework for quantifying predictive uncertainty around model outputs and could therefore provide a useful avenue for developing more informative uncertainty representations in future multi-scale invasion mapping (Angelopoulos and Bates 2022).

More broadly, these findings illustrate how hierarchical multi-scale workflows can bridge scale mismatches in Earth observation. Drone surveys alone are unlikely to support monitoring across landscapes of this extent, while coarser satellite systems may lack sufficient resolution to reliably detect sparse invasion fronts. A targeted UAV calibration layer can link these scales, allowing detailed observations to inform spatially extensive satellite mapping without requiring fine-resolution surveys across the entire landscape. Rather than treating UAV acquisition as a one-off calibration exercise, future recursive approaches could iteratively target surveys towards locations where additional fine-resolution observations are expected to provide the greatest reduction in model uncertainty or constraint on classification error. This could provide a practical optimisation strategy for allocating UAV effort, concentrating surveys in representative locations, areas of particular ecological interest, or regions where model uncertainty is greatest, while reducing the spatial coverage and logistical requirements of systematic drone mapping across the full monitoring area (Holden, Rebelo, and New 2021). Such targeted acquisition may offer a practical compromise between the ecological detail of fine-resolution observations and the spatial coverage required for operational monitoring.

4.3 Detection of sparse *Neltuma* depends on matching sensor grain to canopy structure

Our results show that accurate detection of sparse *Neltuma* depends fundamentally on the correspondence between sensor grain and the spatial structure of the invasion target. Fine-resolution WorldView-2 imagery (1.6 m) enabled useful and operationally relevant mapping of *Neltuma* at scales suitable for district- and protected-area management, whereas detection performance declined progressively at 3 m and then sharply at 10 m as pixel size increasingly exceeded the characteristic scale of sparse canopies and low-density invasion patches. This supports the interpretation that reduced performance at coarser grains reflects a scale-mismatch problem, in which mixed pixels obscure invasion signals and erode sensitivity to low-cover occurrences at invasion fronts.

The steep decline in *Neltuma*-specific accuracy relative to modest reductions in overall classification accuracy further emphasises that conventional accuracy metrics can mask losses in sensitivity to sparse ecological targets. At coarser grains, confusion with *V. erioloba* and other woody vegetation increased as progressive loss of spatial detail reduced discrimination of canopy structure and sub-pixel composition. These findings reinforce the importance of matching observation grain to organismal pattern when using Earth observation for invasive plant detection and extend broader evidence that spatial scale is fundamental to understanding and managing biological invasions (Slade et al. 2023; McMillan et al. 2023; Müllerová et al. 2023). They are also consistent with recent synthesis highlighting spatial resolution as a critical limitation for detecting young trees and sparse stands of *Neltuma* at invasion fronts (Eckert et al. 2025).

An important practical implication is that optical data at 10 m resolution or coarser, including systems such as Sentinel-2 and especially historical archives such as Landsat, are likely to yield highly uncertain detection of sparse and discontinuous invasions. This constrains not only current mapping but also retrospective analyses seeking to reconstruct invasion dynamics from long-term satellite records (Mbaabu et al. 2019). Where fine-resolution imagery is unavailable, improving inference will therefore require approaches that explicitly account for the scale-dependent uncertainty and loss of spatial detail inherent in coarser observations.

More broadly, these results suggest that managing scale mismatch is not simply a technical challenge but a central ecological requirement for reliable invasion mapping (Vaz et al. 2018). Where fine-resolution commercial imagery may be prohibitively costly at national scales, the challenge is therefore not only to acquire finer data, but to develop cost-effective and scalable modelling and monitoring approaches that retain useful information about sparse phenomena at coarser grains. Addressing this remains a key frontier for extending Earth observation approaches to early detection and monitoring of dryland invasions.

4.4 *Neltuma* is expanding across the study area, with broad potential for further densification

The spatial pattern of *Neltuma* distribution strongly supports livestock-mediated dispersal as an important invasion pathway in the western Kalahari, consistent with patterns observed in other semi-arid rangelands (Campos et al. 2011; Kebede and Coppock 2015; Shiferaw et al. 2019). Elevated densities around settlements, roads and dune valleys align with livestock movement patterns, whereby animals concentrate near villages before ranging widely into surrounding savanna (Mosweu et al. 2013; Kamiri, Choge, and Becker 2024; Eschen et al. 2024; Meroni et al. 2017). Since its deliberate introduction in 1980 (van der Poel and Timberlake 1980), *Neltuma* appears to have expanded through coupled processes of local densification and outward dispersal. Although much of the study area remains characterised by low-density infestations, this trajectory suggests substantial potential for continued expansion and densification without effective intervention.

Interactions between livestock sector expansion and episodic drought may further accelerate spread. *Neltuma* is already present in many villages across the western Kalahari (Figure S12), and patterns observed here may foreshadow broader regional dynamics as cattle posts, boreholes and ranches continue expanding westwards (Keeping et al. 2019). Local community accounts of pulse-like invasion during drought support a mechanism in which forage scarcity increases pod consumption and extends livestock ranging into savanna landscapes. Although cattle movement from infested areas into northern Botswana is constrained by foot-and-mouth disease controls (Kabelo et al. 2023), other

531 domestic stock, including goats, sheep, donkeys and horses, may facilitate longer-distance dispersal.
532 Increased livestock mobility, temporary water provisioning and policies promoting grazing expansion
533 risk inadvertently reinforcing invasion processes (Chilume, Keotshepile, and Macinnis-Ng 2025).
534 Current management strategies centred on utilisation are unlikely to provide an effective or
535 economically viable response at landscape scale (Mbaabu et al. 2019). Although Botswana's
536 Integrated Mesquite (*Prosopis* Species) Management Strategy (2022) promotes control through uses
537 such as charcoal and livestock feed, these approaches depend on sufficiently dense infestations to
538 offset harvesting, transport and processing costs. Yet much of the invasion mapped here occurs at
539 low density and across large areas, precisely where utilisation is least economically viable. The
540 BORAVAST community charcoal initiative within the study area illustrates these constraints: *Neltuma*
541 occurred predominantly as extensive, low-density infestations interspersed with relatively small high-
542 density patches, increasing harvesting, transport and kiln deployment costs while limiting economies
543 of scale. This illustrates how invasion pattern, as well as total invaded area, can constrain utilisation-
544 based management in remote dryland landscapes.

545 More fundamentally, utilisation is unlikely to provide a stand-alone solution if the primary
546 management objective is to prevent further spread and densification. The extensive low-density
547 invasion mapped here represents an important window for intervention, but also the stage least
548 suited to biomass-based utilisation. Early-stage removal across remote dune savannas will inevitably
549 be logistically challenging, suggesting a need for approaches that combine targeted control with local
550 knowledge and participation. Local herders may be well positioned to contribute to surveillance and
551 early removal where appropriately supported and remunerated, for example through community-
552 based approaches analogous to Botswana's Herding for Health programme (Heermans et al. 2021).
553 Such approaches may be particularly valuable during early invasion stages, when infestations are
554 widespread but remain relatively sparse and potentially more tractable. More broadly, effective
555 implementation will require a two-way exchange between research, management and local

practitioners, allowing spatial information and ecological evidence to inform action while ensuring that management experience and practical constraints feed back into adaptive strategies (Foxcroft et al. 2020).

Given the escalating costs and declining success of late-stage control (Shackleton et al. 2014), reliance on utilisation alone is unlikely to halt invasion. Instead our results support a spatially differentiated management strategy that prioritises early detection and removal of sparse incursions while targeting dense infestations with context-appropriate control measures within a broader community-based natural resource management framework. Spatially explicit information on invasion density could help prioritise these interventions, distinguishing areas where early removal remains feasible from established infestations where control or utilisation may be more appropriate (Shiferaw et al. 2019). Without sustained and well-resourced intervention, further expansion and densification are likely, with consequences for groundwater availability, livestock production, irrigated agriculture and associated ecosystems, echoing severe social and ecological impacts documented elsewhere in eastern and southern Africa (Ndhlovu, Milton, and Esler 2016; Malongweni et al. 2025; Eckert et al. 2025).

5. Conclusion

This study shows that integrating field observations, drone imagery, and satellite data within a hierarchical multi-scale framework can substantially improve detection and mapping of sparse *Neltuma* invasions, while revealing important limits imposed by sensor grain. Fine-resolution drone data supported accurate plant-scale detection and strengthened satellite classifications through cross-scale calibration, while comparisons among sensors showed that detectability declines sharply when pixel size exceeds the scale of invasion target canopy structures.

For *Neltuma* in the Kalahari, the results indicate substantial potential for continued expansion and reinforce the urgency of proactive, spatially differentiated management while infestations remain relatively sparse. More broadly, our findings demonstrate that targeted integration of fine- and

581 coarse-resolution observations can provide more robust evidence for operational decision-making
582 than either scale alone. For those mapping sparse invasions and other heterogeneous ecological
583 targets, we recommend incorporating fine-resolution UAV observations where the additional
584 ecological detail and classification accuracy can materially improve landscape-scale inference, while
585 using targeted acquisition to maintain operational scalability.

586 As pressures on dryland ecosystems intensify, such hierarchical approaches offer a pathway towards
587 more reliable early detection, scalable ecological monitoring, and stronger environmental intelligence
588 for managing ecological change. Their broader value lies not simply in combining observation
589 systems, but in leveraging their complementary strengths to progressively improve the accuracy,
590 robustness and management relevance of ecological information across scales.

Acknowledgements

This work was supported by the Oppenheimer Programme in African Landscape Systems (OPALS), jointly funded by the University of Exeter, Sarah Turvill, and Oppenheimer Generations Research and Conservation. We thank the Government of Botswana Ministry of Environment, Natural Resources Conservation and Tourism for permission to undertake this research (permit ENT 8/36/4LII(09)). We thank the Civil Aviation Authority of Botswana for authorising the drone surveys. We thank the Dikgosi, Village Development Committees, and the communities of Bokspits, Struizendam and Vaalhoek for their generous support, hospitality and collaboration in the field research. We acknowledge the European Space Agency (ESA) for providing WorldView-2 data through its Third Party Missions programme, and the Norway International Climate and Forest Initiative (NICFI) for providing access to PlanetScope SuperDove imagery. We thank two anonymous reviewers for their constructive and thoughtful feedback, which helped us strengthen the manuscript. For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

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The full codebase for the analysis and resulting outputs including the ground-observed data for training and validation is available at <https://doi.org/10.5281/zenodo.16681147> under a GPL-3.0 license. The processed drone data (georeferenced orthomosaics of spectral reflectance, vegetation indices, and canopy height models) for each of the seven AOIs, and also the output classification maps are available at <https://doi.org/10.5281/zenodo.18506271>.

Disclosure

Our study brings together authors from two countries, including scientists based in the country where the study was carried out. All authors were engaged with the research and study design to ensure that the diverse sets of perspectives they represent was considered. Where relevant, literature

615 published by scientists from the region was cited. Portions of the draft manuscript text were edited
616 with assistance from a large language model (ChatGPT, OpenAI). All outputs were critically reviewed
617 and revised by the authors, who take full responsibility for the content.

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